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React 10m

For a beginner summary of React, return to the page on JavaScript frameworks. In this topic, we'll focus on advanced React features and cover material relevant to your assessments.

An example of a web application using React is a simple counter application available on GitHub. This shopping cart application illustrates CRUD operations. This is an application with an MIT licence, so you're free to use this code. The instructions are given online. The code can be run on GitHub desktop, Cloud or localhost.

Take a look at the counter application and try out the buttons to see how it works. This web application needs a bit more explanation because it's not as simple as just loading an HTML file and viewing it in your browser. You'll notice that it has several folders, but they're not the usual ones like 'img', 'css', or 'js'. Instead, you'll see folders called 'src' and 'public'.

If you try to load the `index.html` file from the 'public' folder, you'll just get a blank white page. That's because the main part of the application is in the 'src' folder. To get it working, you'll need to install several JavaScript files first. These are Node.js modules, which we will explain later, and they're used by Node and React.

You'll also see a 'components' folder inside 'src' with three `.jsx` files. Building web applications with React and similar libraries can feel a bit different from traditional methods, and it might take some time to get used to.

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JSX

As you've seen in the counter application example, we're no longer working with a simple application that uses HTML, CSS and JavaScript separately. React changes how you run the application by introducing a new scripting syntax called JSX, which stands for JavaScript eXtension. Many other frameworks use similar methods, but React is the most widely used one.

In traditional web development, developers keep HTML, CSS and JavaScript in separate files. JSX combines these elements, putting the rendering logic and markup in the same place to create React components. It's important to note that JSX and React are not the same – JSX extends JavaScript syntax, while React provides a library of code.

For example, you can create a button as a React component. You bundle the style, code and markup for the button together in one place. When you need the button, you simply include the button component instead of writing separate code for it across multiple files. This approach makes it easier to reuse and manage elements in your application.

Node.js

Another key aspect of using various JavaScript frameworks and libraries is the use of Node.js. We briefly mentioned this technology earlier, but didn't go into detail. Node.js is often essential for fully utilising these frameworks. Developers frequently use it as a back-end runtime and to build client-side applications with React and other frameworks (Radixweb, n.d.).

Since React is a JavaScript framework, you can integrate it with other JavaScript libraries and code. Node.js includes a package manager called npm (node package manager), which gives you access to over 2 million packages. This is one of the main reasons JavaScript remains one of the most popular programming languages.

By combining React and Node.js, you can create client-side applications as well as applications that work on both the client side and server side.

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Due to the complexity of a Node.js development environment, it deserves an entire topic of its own, which is beyond the scope of this course. The next two frameworks introduced in this topic also rely on Node.js to function effectively. We've included them in this course because JavaScript frameworks are widely used in the industry. However, it's important to have a solid understanding of the core technologies – HTML, CSS and JavaScript – before diving into these frameworks.

Dig deeper

To learn more about React, read these resources:

- *React* – an overview of React's features and what it can do
 - *Quick Start* (React, n.d.) provides information for tutorials, documentation and resources for getting started and learning React
 - *Tutorial: Tic-Tac-Toe* (React, n.d.) provides a tutorial on the basics of using React.



Thoughts on this page?

